

Principles of Aerodynamics

Module 3: Lift, Drag & Airfoil Theory

Academic Year 2025–26 · Prepared by: Dr. Tushar, Dept. of Aeronautical Engineering

1. Fundamentals of Lift Generation

Lift is the aerodynamic force that acts perpendicular to the relative wind and to the span of the wing. It is generated primarily due to the pressure difference between the upper and lower surfaces of an airfoil, a phenomenon explained by both Bernoulli's principle and Newton's third law of motion.

1.1 Bernoulli's Principle

As air flows over the curved upper surface of a wing, it must travel a longer path compared to the air flowing beneath. To maintain continuity, upper-surface airflow accelerates, causing a drop in static pressure. The resulting pressure differential produces an upward net force — lift.

The governing equation (Bernoulli's incompressible form):

$$P + \frac{1}{2} \rho v^2 + \rho gh = \text{constant}$$

where P = static pressure, ρ = air density, v = flow velocity, g = gravitational acceleration, h = height.

1.2 Lift Coefficient (C_L)

The dimensionless lift coefficient quantifies the lifting effectiveness of an airfoil at a given angle of attack (α). It encapsulates the combined effects of camber, planform shape, and flow conditions.

$$L = C_L \cdot \frac{1}{2} \rho v^2 \cdot S$$

L = lift force (N), S = reference wing area (m^2), $\rho v^2 / 2$ = dynamic pressure (Pa).

Key Relationships for C_L

- C_L increases linearly with α up to the stall angle ($\approx 15^\circ$ – 20° for most subsonic airfoils).
- Cambered airfoils generate positive lift even at $\alpha = 0^\circ$.
- Symmetric airfoils produce zero lift at zero angle of attack.
- Beyond the critical angle, flow separates abruptly — stall occurs and C_L drops sharply.

2. Aerodynamic Drag

Drag is the aerodynamic resistance force acting parallel to and in the direction of the relative wind. In level flight, the engine thrust must exactly overcome total drag for steady-state conditions.

2.1 Components of Total Drag

Drag Component	Also Known As	Description
Parasite Drag	Zero-lift drag	Skin friction + form drag; independent of lift
Induced Drag	Drag due to lift	Arises from wingtip vortices; decreases with speed

Wave Drag	Compressibility drag	Occurs near/above Mach 1 due to shockwave formation
Interference Drag	Junction drag	Caused by airflow interaction at airframe junctions

Table 1 — Classification of aerodynamic drag forces.

3. Airfoil Geometry & Nomenclature

An airfoil (aerofoil) is the cross-sectional shape of a wing, blade, or sail designed to produce an aerodynamic reaction when moved through a fluid. Understanding its geometric parameters is essential before analysing performance.

3.1 Key Geometric Parameters

Parameter	Symbol	Definition
Chord Length	c	Straight-line distance from leading edge to trailing edge
Mean Camber Line	—	Locus of points midway between upper and lower surfaces
Maximum Camber	m	Greatest distance between chord line and mean camber line
Maximum Thickness	t	Greatest distance between upper and lower surface (% chord)
Angle of Attack	α	Angle between chord line and the relative wind direction
Leading-Edge Radius	r	Radius of curvature at the foremost point of the airfoil

Table 2 — Standard airfoil geometric parameters.

3.2 NACA Four-Digit Series

The NACA (National Advisory Committee for Aeronautics) four-digit series provides a systematic notation for airfoil geometry. For example, **NACA 2412** is decoded as follows:

- **2** — Maximum camber of 2% of chord length
- **4** — Location of maximum camber at 40% of chord from the leading edge
- **12** — Maximum thickness of 12% of chord length

4. The Lift-to-Drag Ratio (L/D)

The lift-to-drag ratio is the most important measure of aerodynamic efficiency. A higher L/D ratio means the aircraft generates more lift per unit of drag, translating to better range and fuel economy.

For commercial jet transports, typical cruise L/D values range from **15 to 20**. High-performance gliders can achieve L/D ratios exceeding **60**. The maximum L/D $(L/D)_{\max}$ occurs at the angle of attack where induced drag equals parasite drag.

$$(L/D)_{\max} = C_L^{1/2} / C_D \text{ (at optimum } \alpha \text{)}$$

Review Questions

- Q1.** Explain the physical mechanism by which an airfoil generates lift, using both Bernoulli's principle and Newton's third law.
- Q2.** A wing of area 25 m² flies at 90 m/s at sea level ($\rho = 1.225 \text{ kg/m}^3$) with $C_L = 0.8$. Calculate the lift force.
- Q3.** Sketch a C_L vs α curve for a cambered airfoil. Indicate the stall angle and the zero-lift angle of attack.
- Q4.** Differentiate between induced drag and parasite drag. How does each vary with flight speed?

Q5. Decode the NACA 4415 airfoil designation and state its geometric properties.

Next Module — AE 202: Propulsion Systems & Jet Engine Thermodynamics